MyLearna
Plan • Capture • Grow

MATHEMATICS

Name: ____________________

Date: ____________________

Use Mathematical Modelling in Financial and Real-World Contexts

Step 40

⭐ Today I am learning to use mathematical modelling to solve financial and real-world problems.

1. Model and Solve

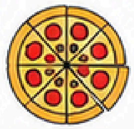
Read each problem. Create a model (table, diagram or equation) to solve it. Show your working.

a) Sarah buys 3 notebooks at $2.75 each and a pen that costs $1.80. She pays with $20. How much change does she get?

Model:

Answer: $ __________

b) A pizza is cut into 8 equal slices. Jason eats 3 slices and Mia eats 2 slices. What fraction of the pizza is left? What percentage is left?



Model:

Fraction left: __________ Percentage left: __________%

c) A movie ticket costs $12.50. A family of 4 goes to the movies and buys a large popcorn for $8.75 and 2 drinks at $3.25 each. What is the total cost?

Model:

Answer: $ __________

d) Emma saves $15 each week in her savings jar. After 6 weeks, she spends $32 on a gift. How much money does she have left?

Model:

Answer: $ __________

2. Budgeting

Use the table to plan a weekly budget. Decide how much money will be spent and how much will be saved.

Item	Planned Spend ($)	Actual Spend [illegible]	Difference (Actual – Planned)
Groceries	60.00	58.75	
Transport	25.00	27.40	
Lunch / Snacks	20.00	18.60	
Entertainment	15.00	21.30	
Savings	30.00		

a) Complete the Difference column.

b) Did you spend more or less than planned? By how much? $ __________

c) How much money remains for savings? $ __________

3. Real-World Modelling

Write an equation or model to represent each situation. Then solve.

a) A plant costs $18.50. The price increases by 12%. What is the new price?

Model / Equation:

Answer: $ __________

b) A shop has a 20% discount on a jacket that costs $85. What is the sale price?

Model / Equation:

Answer: $ __________

c) A car uses 7.5 litres of fuel to travel 100 km. How much fuel will it use to travel 560 km?

Model / Equation:

Answer: __________ litres

d) A box of chocolates has 24 pieces. If you eat $\frac{1}{6}$ of them, how many are left?

Model / Equation:

Answer: __________ pieces

4. Challenge

Create your own real-world situation that involves money, percentages or rates. Write a model (table, diagram or equation) and solve it.

My situation: ____________________

My model:

My answer:

My Learning Check

Circle a face.

- I can do it!
- Almost there!
- I need more help.
- This was hard.

Parent Observation (optional)

- ☐ Understood the concept
- ☐ Created correct models
- ☐ Solved problems accurately
- ☐ Used appropriate operations
- ☐ Neat and organized work
- ☐ Worked independently
- ☐ Needed some support
- ☐ Demonstrated understanding verbally

Notes: ____________________